

1 BDE Log Records

1.1 General

The "posting" function generates log records from recorded events. Log records describe a period of time and include the evaluation of different values, such as quantities, durations or performances.

A BDE log record is an evaluated data record that is generated due to posting events. "Evaluated" means that this data record

- generally refers to a period of time (in an extreme case, this period of time is reduced to a point in time),

and

- includes durations (machine times, labor utilization, resource performance accounts) or
- evaluated quantities (e.g. yield, scrap)

HYDRA-BDE log records describe postings based on operations, staff and batches. Which log record is generated at which point in time depends on different factors, such as triggering posting events, configured posting rules or the HYDRA modules in use. The different types of log records are differentiated by their record type. The following record types based on orders are distinguished:

1.2 Record type A

General

A log record of the record type A is generated when an operation is logged on. It has been designed to be able to upload operation logons to the higher-level ERP/PPS system.

Triggering events and dialogs

Events: Logon of operation (A_AN)

Dialogs: Log operation on (A_AN),

Log operation and person on (together) (A_P_AN),

Log operation on with output batch (A_AN_MPL),

Beginning of shift (A_AAN)

etc.

Special remarks

Using optional configurations, different other dialogs can also cause an operation to be logged on (e.g. change order if the machine status changes).

The log record is not displayed in the maintenance of postings function. Consequently, it cannot be edited (changed, deleted).

1.3 Record type T

General

A log record of the record type T is generated when "partial uploads/confirmations" are recorded. In contrast to all other record types that refer to a period of time, this log record always refers to exactly one point in time (beginning of posting and end of posting are identical in the log record). Thus, this log record does not include durations, i.e. fields relating to durations do not include a value (0). The log record contains the quantities recorded with the upload.

Triggering events and dialogs

Events: all events that include a manual quantity (A_TR, A_UN, A_AB, etc.)

Dialogs: all dialogs that include a manual quantity .

Special remarks

In general, a partial upload may be performed for an operation that is logged on. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded by way of a "quantity upload".

All quantities posted onto an operation are included in the single partial uploads (record type T) and as total in the log records "interruption of operation" (record type U) and "logoff of operation" (record type E). Thus, a T record is generated for each quantity that is collected and posted on an operation.

Manual quantities may be recorded in any dialogs, e.g. in a partial upload, when personnel is logged off, when an operation is interrupted or logged off, when batches are posted, etc. The log record of the record type T with manual quantities is directly generated, when the dialog is recorded/posted (not only when the operation is logged off/interrupted, for example).

In addition, the automatic collection of counters allows for quantities to be posted onto active operations. T records are generated for these quantities as well. HYDRA accumulates the quantities from the automatic collection over a period of time and generates the corresponding T record(s), **but only if the scenario changes**. This may be, for example, an order or person that is logged on manually, a manual, partial upload/confirmation or a manual or automatic machine status change.

The single partial uploads/confirmations also represent the collection of quantities with different reasons (e.g. scrap reasons).

Example:

Scrap (quantity 1) with the reasons 1, 2 and 3 each was recorded for an operation. The U or E record shows a total scrap quantity of 3. In addition, HYDRA records three T records each with a scrap quantity of 1 and the corresponding reason.

1.4 Record type U

General

A log record of the record type U is generated, when an operation is interrupted. Either the user can interrupt the OP manually or it can also be made automatically, for example, by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record of an operation includes the period of time between logon and interruption of the operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the operation was logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities and times posted in the log record are also posted onto the operation status.

Triggering events and dialogs

Events: Interruption of operation (A_UN),

Quantity upload to an inactive operation (A_MR)

Dialogs: Interruption of operation (A_UN),
 Quantity upload to an inactive operation (A_MR)
 End of shift (A_AUN)
 etc.

Special remarks

In general, an operation may only be interrupted if it was logged on before. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded using a "quantity upload". In this case, the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be interrupted even by different other dialogs through optional configurations (e.g. order change when the machine status is changed or automatic interruption of the operation when the target quantity is reached).

1.5 Record type E

General

A log record of the record type E is generated, when an operation is logged off. As regards content, this data record corresponds to a log record of the record type U.

Triggering events and dialogs

Events: Operation is logged off (A_AB),
 An inactive operation is terminated (A_BE)

Dialogs: Operation is logged off (A_AB),
 An inactive operation is terminated (A_BE)
 etc.

Special remarks

If an inactive operation is finished the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. the operation is logged off automatically, when the target quantity is reached or predecessor operations are logged off automatically).

1.6 Record type H

General

A log record of the record type H is generated, when an (output) batch is changed/logged off (ADE-CHV, MPL). Either the user can change the batch manually or it can also be made automatically, for example, by a machine signal or by interrupting the operation or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between logon and logoff of the batch to/from the machine and operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the batch and operation were logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities posted in the log record are also posted onto the batch.

Triggering events and dialogs

Events: Logoff of (output) batch (CA_AB)

Dialogs: (Output) batch change (CA_WL)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN) etc.

Special remarks

A batch posting does neither include labor utilization nor personal resource performance accounts.

A batch may also be triggered to be posted even by different other dialogs through optional configurations.

1.7 Record type B

General

A log record of the record type B is generated if a person is logged off from a workplace and an operation that is active at this workstation. Either the user can log the person off manually or it can also be made automatically, for example, by interrupting the operation or by the person clocking out (HYDRA-PZE: out) or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between the person being logged on and logged off to and from the workplace and operation. The durations included in the log record refer to exactly this period of time and, subject to the respective configuration, they are compared with the shift model of the workplace to which the person and operation were logged on or with the person's BDE shift model: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

Triggering events and dialogs

Events: Log person off (P_AB)

Dialogs: Person is logged off (P_AB)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN)
 HYDRA-PZE out (P_GEH)
 etc.

Special remarks

A person may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. staff is logged off when the machine status is changed).

1.8 Special features

Evaluation date

The field "evaluation date" is defined for the work day evaluation of PZW (Personnel Time Management) and the wage calculation of the LLE module. It has been designed to assign postings to a settlement day for the LLE module and for the BDE/PZW comparison. Due to flexible working times, this settlement day may differ from the ADE shift date, in particular for night shifts.

Further information can be found in the document entitled [GLOSSARY_EvaluationDate.pdf](#).

Corresponding shift

The time stamp for logging off pertaining to a BDE log record determines the shift (shift date, shift number) to which the log record belongs.

Operation splits

When it comes to split operations, the log record is ONLY generated for the split operation. A log record is not generated for the split master itself. The status of the split master is only updated.

Merged operations

Log records are generated for merged operations as well as for individual operations. The statuses of individual operations are also updated accordingly. While posting, the recorded quantities and times of a merged operation are distributed to the corresponding individual operations according to different configurations. The log records of individual operations only include proportionate quantities and times.